| Civics Group | Index Number | Name (use BLOCK LETTERS) |
|---|---|---|
| | | |

**H2**

**ST ANDREW'S JUNIOR COLLEGE**
**2015 JC2 Preliminary Examinations**

**H2 BIOLOGY** **9648/3**

# Paper 3: Applications and Planning Question (Mark Scheme)

Wednesday 16 September 2015 2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

**READ THESE INSTRUCTIONS FIRST**

Write your civics group, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,
1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **Paper 3** | |
| **1** | /14 |
| **2** | /13 |
| **3** | /13 |
| **Total** | **/40** |
| | |
| **4 (Planning)** | **/12** |

This document consists of **14** printed pages.

**[Turn over**

*Answer* **all** *questions*.

**1.** Patients suffering from Life-threatening Multiple Thrombosis (LMT) are unable to synthesise sufficient amounts of platelet factor IV (PF4), which are released by activated platelets at sites of injury to promote blood coagulation. Platelets are ubiquitous in the blood stream.

Recent advances in genetic engineering have seen preliminary success in cloning human PF4 gene into bacteria for the mass production of PF4 to be used in clinical treatments of LMT.

Fig. 1.1 shows the simplified map of the plasmid, pYH29, used in the cloning of PF4 gene. The selectable markers used are genes that confer resistance to two antibiotics: Kanamycin (KanR) and Clindamycin (ClnR).

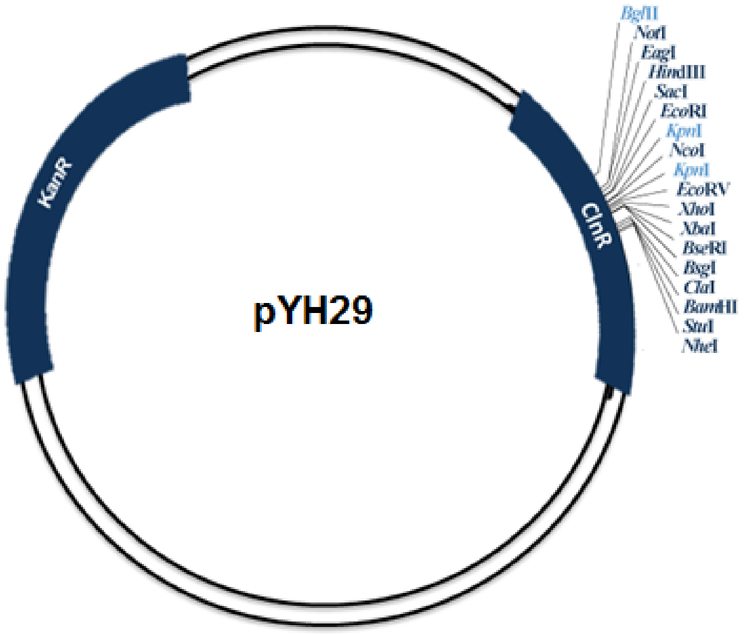


**Fig. 1.1**

**(a)** Describe the natural function of a restriction enzyme such as *HindIII.*

……………………………………………………………………………………..………………..…..[1]

1 Cleave any <u>foreign DNA</u> that enters the bacterial cell,
protecting <u>bacteria</u> from attack by bacteriophages / virus.

**(b)** The amount of PF4 gene inserts obtained from cDNA library is normally too little for successful cloning. Suggest one way to increase the amount of the PF4 gene inserts.

……………………………………………………………………………..…………………..........[1]

1 Perform <u>PCR</u> to <u>amplify</u> the PF4 gene inserts.

**(c)** The PF4 gene insert was obtained from cDNA library and cut with *Hin*dIII (restriction enzyme). Describe and explain the subsequent procedures for cloning the PF4 gene into *E. coli* cells.

……………………………………………………………………………………………………[4]

1 Plasmid pYH29 is cut with HindIII to **generate complementary sticky ends**.
2 Plasmid and PF4 gene insert are mixed, **anneal** with each other by forming hydrogen bonds between complementary sticky ends
3 DNA ligase added to seal the nicks by formation of phosphodiester bonds between adjacent nucleotides (on the sugar phosphate backbone)
4 *E. coli* cells made competent by addition of $Ca^{2+}$; transformation is carried out via heat shock treatment.

**(d)** Using both antibiotics, explain how the researchers ensure that only colonies which contain the PF4 gene insert are cultured in large quantities.

……………………………………………………………………………………………………[5]

~~1~~ Plated bacteria on **LB/Kan plate** to select for **transformed bacteria/bacteria that have taken up the plasmid;**
2 Replica plated on **LB/Cln** plate; comparison of bacterial colonies on LB/Kan plate and LB/Cln plate.
3 Colonies that cannot survive on LB/Cln plate contain recombinant plasmid; Insertion of PF4 gene disrupts ClnR gene;
4 Perform **gene probing** using DNA probes that is **complementary to PF4 gene.**
5 Ref. X-ray autoradiography

**(e)** The PF4 gene, when introduced into E. coli host cells, produces PF4 that is biologically inactive. Explain why and suggest a possible way to overcome this problem.

……………………………………………………………………………………………………[2]

1 cDNA inserted in incorrect orientation, resulting in transcription of incorrect strand, hence, encoding for a different polypeptide
2 Use different restriction enzymes to create different sticky ends on each end of the PF4 gene and plasmid.

OR

3 Eukaryotic post-translational mechanisms not present in prokaryotic cells
4 Use a mammalian host cell for cloning of PF4 gene.

**(f)** Identify one additional feature of pNY29 plasmid (not shown in Fig. 1.1) and explain why it is needed for the successful production of PF4.

……………………………………………………………………………………………………[1]

1 Prokaryotic promoter; allows recognition by host cell's RNA polymerase for expression of inserted PF4 gene in bacteria cell
2 Origin of replication; recognisable by prokaryotic enzymes to enable independent replication of the plasmid and foreign gene inside the host cell.

**[Q1 Total: 14]**

**2(a).** X-linked haemophilia is a disease that impairs the body's ability to control blood clotting. The gene affected normally produces Factor VIII, a component of the blood coagulation pathway.

A RFLP marker that is linked to X-linked haemophilia is discovered. Fig. 2.1 shows the two RFLP markers with their restriction sites.

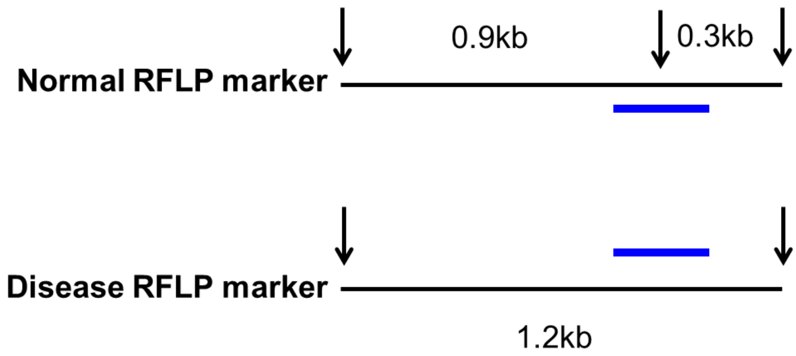


**Fig. 2.1**

PCR was carried out to amplify the target DNA segment containing the RFLP allele. After which, restriction digestion using *BclI* was done followed by gel electrophoresis to obtain the results shown in the PCR-RFLP analysis in Fig. 2.2. The results are accompanied by the pedigree of a family affected by the disease.

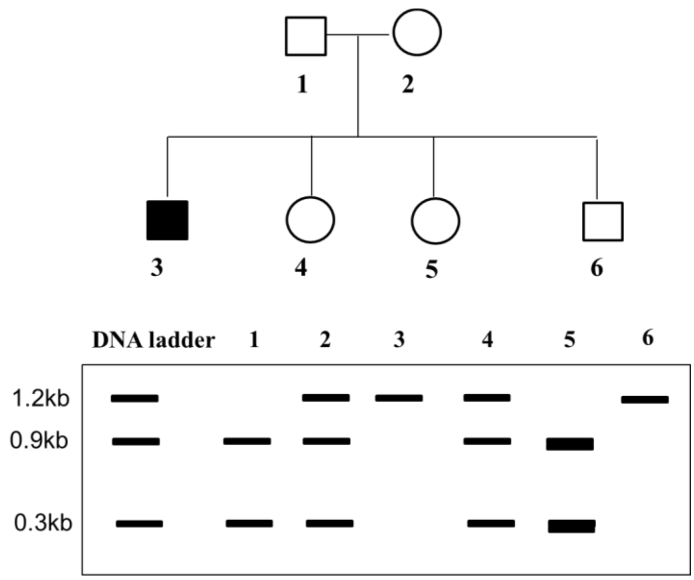


**Fig. 2.2**

**(i)** State the role of the DNA ladder seen in Fig. 2.2.

………………………………………………………………………………………………….[1]

1 as a reference/standard
/ providing known DNA fragments of specific lengths;
to estimate the size/number of repeats of unknown DNA molecules;

**(ii)** On Fig. 2.1, draw the location of the probe on both alleles that would produce an autoradiogram as indicated in Fig. 2.2.
……………………………………………………………………………………..………..….……[1]

**(iii)** Contrast and explain the banding patterns of individuals **1** and **5** as seen in Fig. 2.2.
……………………………………………..……………………………………………..….……[2]

1 [Difference] Individual 5 has thicker bands than individual 1;;
2 [Explanation] Individual 5 has two copies of X chromosome while individual 1 has only one copy of X chromosome;;

**(iv)** Other than mutations, account for the observations for individual **6** as seen in Fig. 2.2.
……………………………………………………………………………………..………..….……[3]

1 [Observation] Individual 6 has the diseased RFLP marker but he does not have haemophilia, as shown by the 1.2kb fragment;;

2 [Explanation] Crossing over occurs in meiosis 1 during the formation of gametes in the **mother / individual 2**;;

3 Resulted in the formation of a recombinant gamete where the disease RFLP marker is no longer linked to the disease allele;;

**(b)** Suggest why the disease-causing allele causing certain diseases can only be mapped using linkage analysis instead of probing for the disease-causing allele directly.
……………………………………………..……………………………………………..….……[1]

1 Nucleotide sequence of the disease-causing allele is not known;;
/ Mutation in the disease-causing allele in the genome is not known;;
/ The mutation in the disease-causing allele did not introduce or remove a restriction site;;

Reject: There are no restriction sites in the disease-causing allele – unlikely.

**(c)** DNA profiling is a forensic technique used to identify individuals by characteristics of their DNA.

Explain the basis of application of VNTR in DNA profiling.
………………………………………..………………………………………..…..……[3]

1 Each individual has variable number of VNTR repeating units at the same locus on each homologue
/ The number of repeating segments varies greatly from individual to individual;;
2 VNTR regions are flanked by restriction sites on either side which are cut by RE;;
3 Gives rise to fragments of different lengths which can be separated by gel electrophoresis;;
4 Analysis of multiple VNTRs / VNTR at **different loci** (consisting of different repeat sequences) will produce **different band patterns** that gives each individual unique DNA fingerprint;;

Another method to detect ADA-SCID is through the use of PCR. Fig 2.3 shows the number of DNA molecules made using PCR, starting with one molecule.

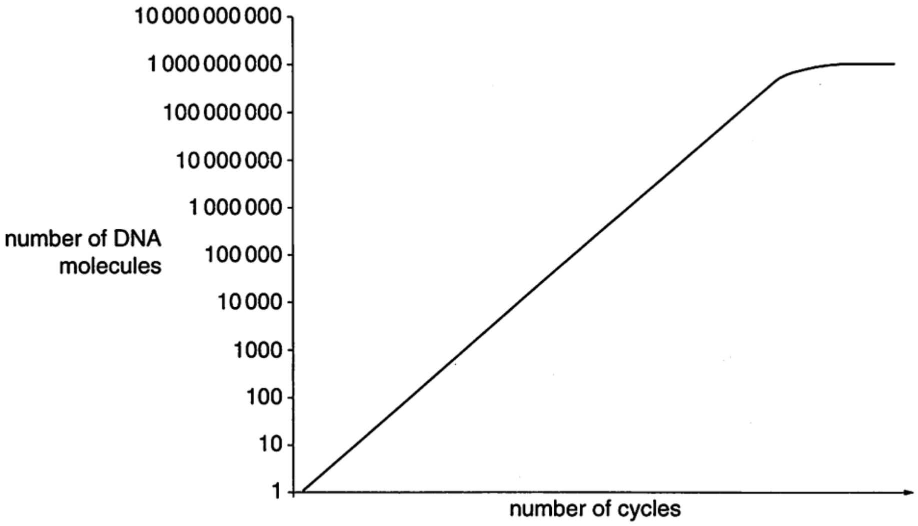


**Fig. 2.3**

**(d)** Suggest two reasons for the leveling off of the graph from cycles 17 to 20.
……………………………………………………………………………………..……[2]

1 Nucleotides are used up (hence is unable to make complementary chains)
2 Primers are used up
3 the template DNA strands anneal with each other instead of with the primer
4 *Taq* DNA polymerase gradually becomes denatured ref. no elongation

**[Any 2]**

**[Q2 Total: 13]**

**3(a)** Golden rice is a variety of rice (*Oryza sativa*) produced through genetic engineering to produce β-carotene in the edible parts of rice, the endosperm.

Golden rice was created by transforming rice callus tissue with a DNA construct consisting of two β-carotene biosynthesis genes and an antibiotic resistance gene. The genes were placed under the control of a tissue-specific promoter, Glu.

- *psy* (phytoene synthase) from daffodil (*Narcissus pseudonarcissus*)
- *crtI* (carotene desaturase) from the soil bacterium *Erwinia uredovora*
- *Hyg resist* – antibiotic resistance gene from a bacterium

**(i)** Explain why rice has been genetically modified to produce extra β carotene.

…………………………………………………………………………………………………[2]

1 vitamin A deficiency in developing countries
2 prevention of blindness
/ reduces susceptibility to diseases
3 rice is a staple food / forms a major part of diet (in those countries) ;

**(ii)** Explain what is meant by callus tissue.

…………………………………………………………………………………………………[1]

mass of disorganised / undifferentiated / unspecialised plant cells

**(iii)** Suggest the role of the tissue-specific promoter, Glu.

…………………………………………………………………………………………………[2]

1 To allow the genes to be expressed / switch on genes
2 only in the endosperm / edible parts of rice, prevent wastage of resources

**(iv)** Explain the role of the Hyg resistance gene in this procedure.

…………………………………………………………………………………………………[1]

To enable selection of successfully transformed cells
/ Only the successfully transformed cells can grow in the presence of the antibiotic;

**(b)** The original choice of a *psy* gene from daffodils was made because daffodils produce large amounts of β-carotene in their yellow petals, and because they are monocotyledonous plants, like rice.

In the subsequent years following the invention of Golden Rice, investigations were carried out to see if *psy* genes taken from species other than daffodils would produce greater quantities of β carotene than the first types of Golden Rice.

- *Psy* genes were isolated from the DNA of carrots, daffodils, maize, peppers and tomatoes.
- The genes were inserted into different plasmids.
- The promoter Glu, and *crtl* genes from *E. uredovora*, were also inserted into all of the plasmids.
- The five types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The % carotenoid content produced by these rice cells was measured.

Fig. 3.1 show a bar graph of the carotenoid content of calluses transformed with *psy* gene from the five different sources.

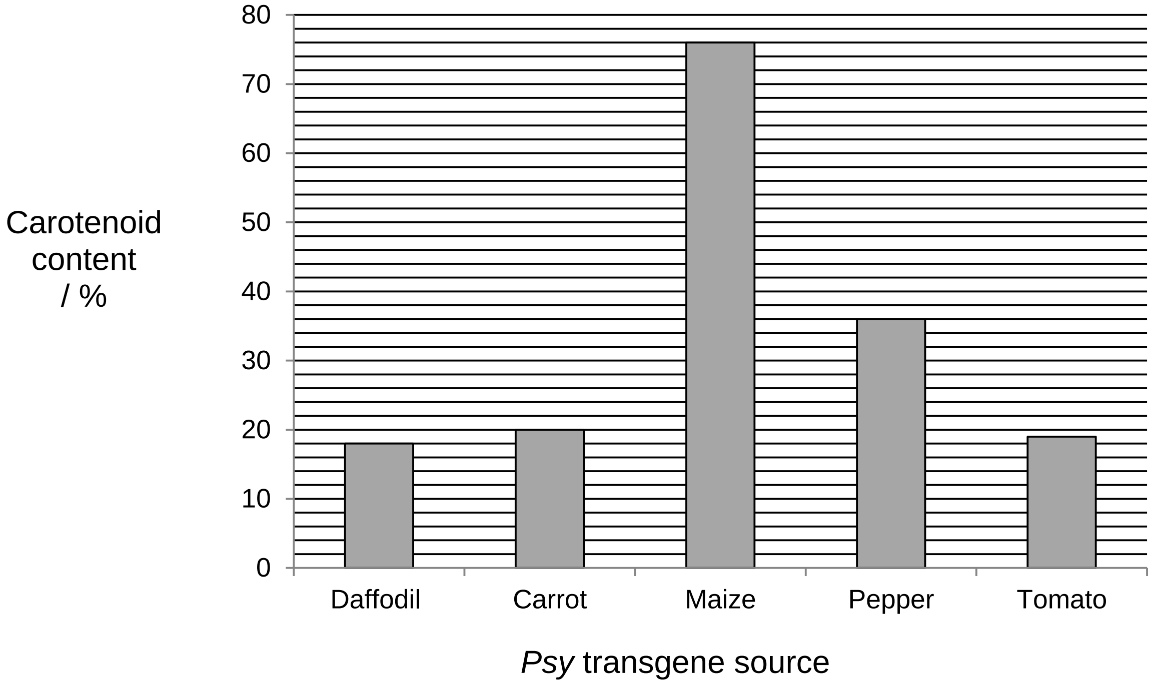


**Fig 3.1**

**(i)** Contrast the effectiveness of the *psy* genes of various species in increasing the carotenoid content over that of the daffodil.

…………………………………………..…………………………………………..…..…..[2]

<u>Any 2 points</u>

1 *Psy* gene from maize gives the <u>largest</u> increase (<u>58%</u>) in carotenoid content.
2 *Psy* gene from pepper increases the carotenoid content (<u>40%</u>).
3 *Psy* gene from tomato and carrot gave <u>slight</u> increase in carotenoid content (<u>1 and 2%</u>, respectively).

**(ii)** Suggest reasons for the differences in carotenoid content of the calluses.

……………………………………………………………..……………………………........…[3]

1 different DNA sequences in the *psy* genes from different sources
2 different amino acid sequences, different tertiary structure in the enzyme
3 which will affect the interaction/affinity of enzyme (phytoene synthase) with its substrate / co-factors / other proteins

**(iii)** Describe the possible social implications of growing Golden Rice.

……………………………………………………………..……………………………........…[2]

<u>Any 2 points</u>

1 unfair for large multinational companies to patent GM plants / may benefit profit-driven research companies more than farmers/consumers in underdeveloped nations.
2 Increased dependence of undeveloped countries on rich developed countries, (might reduce efforts to relieve poverty)
3 GM seed could be difficult for farmers in developing countries to obtain
4 high cost of buying (new) GM seed / cannot use own seed

**[Q3 Total: 13]**

**4. Planning question**

You are required to plan, but not carry out, an investigation into the effect of carbon dioxide concentration on the rate of photosynthesis in an aquatic plant, *Elodea canadensis.*

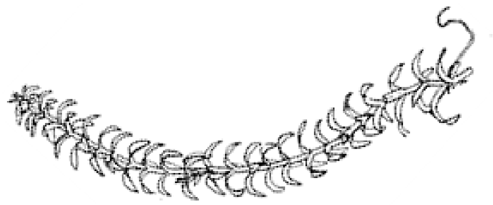

**Fig 4.1**

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- pieces of the *Elodea* plant, each with varying lengths exceeding 100 mm long shown in Fig. 4.1
- plasticine (to anchor *Elodea* plant in water)
- an unlimited supply of filtered pond water through which air has been bubbled for at least 24 hours
- an unlimited supply of 0.1 mol $dm^{-3}$ sodium hydrogen carbonate solution
- boiling tubes
- bungs to fit the test-tubes with one L-shaped glass tube passing through
- pieces of capillary tube each 300 mm long, with a bore diameter of 1 mm
- beaker of coloured liquid
- rubber connecting tubings to connect L-shaped glass tubes or capillary tubes
- retort stand with clamps
- scalpel
- ruler marked in mm
- stop watch
- thermometer
- large glass beaker
- supply of hot and cold water
- a lamp at one end of an otherwise unlit bench 2 metres long
- a room which can be made dark
- thick sheet of clear glass (heat shield)

Your plan should:

- show an explanation of the theory to support your practical procedure
- describe the method with scientific reasoning so that the results are as accurate and reliable as possible
- be illustrated by relevant diagrams, if necessary
- show how you will record your results and the proposed layout of results table and graphs,
- include reference to safety measures to minimise any risks associated with the proposed experiment,
- use the correct technical and scientific terms.

**[Q4 Total: 12]**

**MARK SCHEME**

**1. Background theory**

- Light is absorbed by pigments in the photosystems / Photoactivation of chlorophyll at the reaction centre occurs and chlorophyll becomes oxidised.
- Photolysis of water results in the production of $O_2$
- $CO_2$ is needed in the Calvin cycle where it is used to make sugar.

**2. Hypothesis**

- As the concentration of $CO_2$ increases the rate at which carbon is incorporated into carbohydrate (Accept: the rate of photosynthesis increases)
- At high $CO_2$ concentration, further increases in the $CO_2$ concentration will have no effect on rate of photosynthesis
- another factor becomes limiting

**[**2 pts: **½ mark,** 3 pts: **1 mark,** 4 pts: **1 marks,** 5-6 pts: **2 marks]**

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**3. Variables [2 marks]**

Independent variable
- Concentration of hydrogen carbonate solution (eg. 0.02, 0.04, 0.06, 0.08, 1.00 mol $dm^{-3}$)

Dependent variable
- Rate of photosynthesis, measured by volume of $O_2$ consumed per min

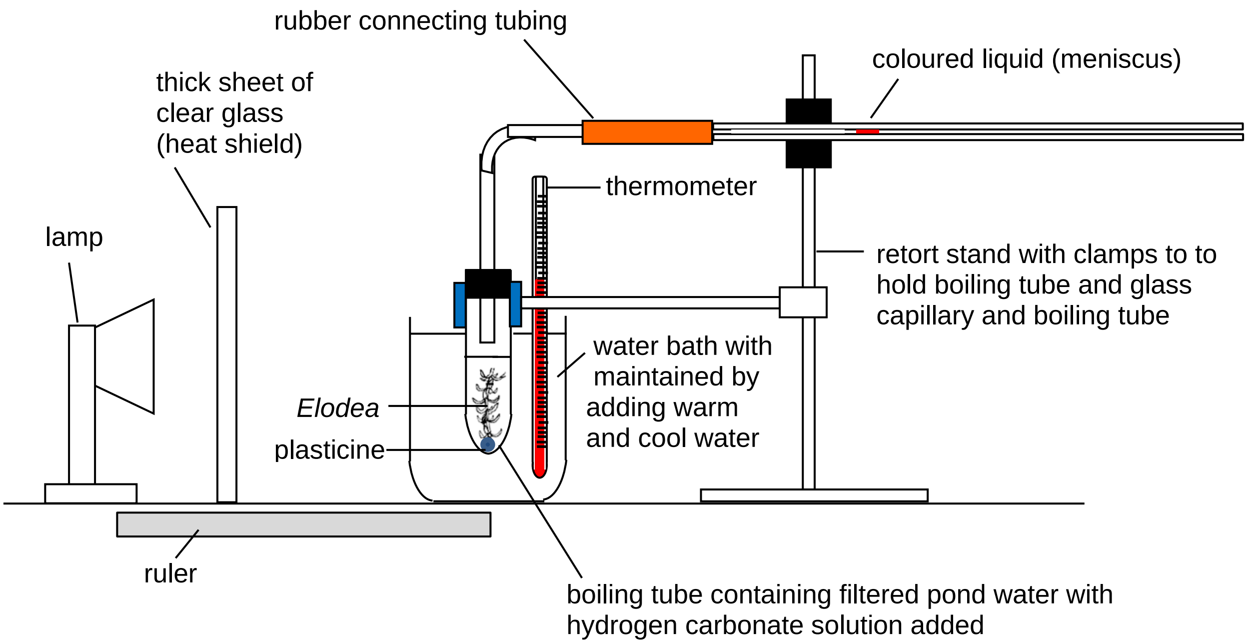
Other variables to be kept constant (any 2 variables)
- Same light intensity; use same light source at same distance from plant (specified, eg. 30 cm)
- Same length of *Elodea* (specified, eg. 5 cm); ensure that each experiment is being carried out with the same leaf surface area
- Same temperature (specified, eg. 25°C); block out heat from lamp using glass sheet

**4. Control with justification [1 mark]**
- Same length of rubber tubing (inanimate object) to replace *Elodea,* (control being subjected to same environmental factors as that for the experiment)
- Purpose of control - To ensure that only *Elodea* plant is giving out $O_2$ during experiment

OR
- Place the same setup in the dark (control being subjected to same environmental factors as that for the experiment)
- Purpose of control - To calculate the amount of $CO_2$ evolved during respiration so as to calculate the actual amount of $O_2$ evolved during photosynthesis

OR
- No hydrogen carbonate solution added (control being subjected to same environmental factors as that for the experiment)
- Purpose of control - To calculate the amount of baseline $O_2$ evolved during photosynthesis

## 5. Procedure




- Description of apparatus setup / labelled diagram
  **[**4-5 pts: **½ mark,** 6-7 pts: **1 mark]**

  - Lamp and heat shield
  - Ruler to measure distance between lamp and *Elodea*
  - Water bath with thermometer
  - *Elodea* anchored by plasticine submerged in filtered pond water with hydrogen carbonate solution added
  - Connecting tubes (should not be submerged)
  - Coloured liquid in capillary tube
  - Retort stand and clamps

**[2 marks]**

- Dilution table showing a range of hydrogen carbonate solutions of known concentration (eg. 0.02, 0.04, 0.06, 0.08, 1.00 mol $dm^{-3}$)

- Specified time for equilibration (Accept: 2-10 min), ensure that drop of coloured liquid is moving smoothly in the capillary tube before starting measurement.
- Reason: Allow plant to acclimatise to light and hydrogen carbonate solution in test tube

- Specified time for measurement of distance moved by coloured liquid using ruler (Accept: 5-10 min)

**6. Results table with appropriate headings**

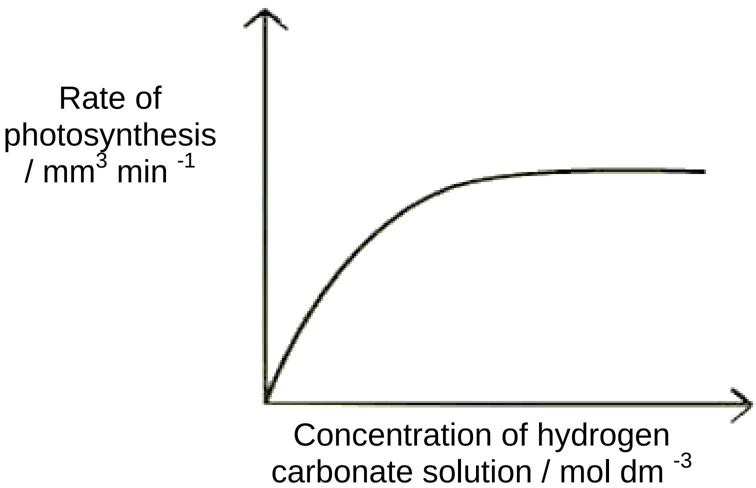
- Volume of $O_2$ given out = $\pi r^2$ x Distance travelled by coloured liquid **[½ mark]**

**[½ mark]** **[½ mark]**

| Concentration of hydrogen carbonate solution / mol $dm^{-3}$ | Distance travelled by coloured liquid in 5 minutes / mm | | | | Volume of $O_2$ given out / $mm^3$ | Rate of $O_2$ given out / $mm^3$ $min^{-1}$ |
|---|---|---|---|---|---|---|
| | Reading 1 | Reading 2 | Reading 3 | Average | | |
| 0 | | | | | | |
| 0.02 | | | | | | |
| 0.04 | | | | | | |
| 0.06 | | | | | | |
| 0.08 | | | | | | |
| 0.10 | | | | | | |

**Theoretical graph with correct labels and units [½ mark]**

- Theoretical graph of rate of photosynthesis OR rate of $O_2$ given out / $mm^3$ $min^{-1}$ against concentration of hydrogen carbonate solution / mol $dm^{-3}$




**7. Reliability [1 mark]**

- Bung and test tube must be airtight / rubber connecting tubing and capillary tube must be airtight
- 3 replicates for each hydrogen carbonate concentration
- 3 repeats using fresh materials and new set-up

**8. Risk and Safety [1 mark]**

- Scalpel is sharp and may cause cuts; handle carefully to when trimming stems
- Glass capillary tube is fragile and sharp when broken and may cause cuts; handle and dispose broken pieces carefully
- Lamp is hot when in use; handle carefully to prevent scalding with bulb
- Use dry hands when operating the lamp to prevent electric shock

## Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**5** **(a)** There are two common forms of severe combined immunodeficiency (SCID). Contrast between adenosine deaminase (ADA) deficiency and X-linked SCID. [5]

**(b)** Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]

**(c)** Discuss the goals and benefits of the human genome project. [7]

**[Total: 20]**

**5(a) There are two common forms of severe combined immunodeficiency (SCID). Contrast between adenosine deaminase (ADA) deficiency and X-linked SCID. [5]**

| **Point of comparison** | **Adenosine deaminase (ADA) deficiency** | **X-linked SCID** |
|---|---|---|
| 1. Type of genetic disorder | Autosomal recessive disorder; | Sex-linked recessive disorder; |
| 2. Gene involved | Mutation in the ADA gene; | Mutation in the common gamma chain gene; |
| 3. Location of the gene | Found on chromosome 20; | Found on the X-chromosome; |
| 4. Resultant protein coded for | Non-functional ADA/ no ADA produced; | Non-functional common gamma chain (interleukin receptor)/ no common gamma chain (interleukin receptor); |
| 5. Effect on the immune system | B and T lymphocytes are killed due to accumulation of deoxyadenosine; | B and T lymphocytes fail to proliferate and differentiate; |

**5(b) Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]**

1 Named example: blood stem cells;
Differentiate into cells of the same lineage;
E.g. red blood cells, white blood cells and platelets;
2 Replace cells worn out from normal wear and tear;

3 (blood stem cells) isolated from bone marrow (of patient);
No cell rejection;
Cultured *in vitro* first to **increase numbers**;

4 Attenuate retroviral vector by removing viral genes that cause harm (eg. lysis);
5 mRNA of target gene inserted in vector;
ref. ADA gene / IL2RG gene;

6 Recombinant vector allowed to infect cultured blood stem cells;
Viral genome is reverse transcribed;
and resulting cDNA is then integrated into host cell DNA;
7 can be **stably propagated** by chromosomal replication following cell division

8 Selection of transformed blood stem cells on selection plates;
**Transformed** blood stem cells grown in culture to **increase number** of cells
Transgenic blood stem cells grown reimplanted/injected into patient;

9 ref. normal functional protein product, thereby restoring the correct function of the target cells and altering the phenotype
functional **T and B lymphocytes** in the patient;
to fight off infections / development of functional immune system;

**5(c) Discuss the goals and benefits of the human genome project. [7]**

**[Goals] [half mark per point; 3m]**

1 Identify all the <u>genes</u> in human DNA (ref. 30,000 genes)
2 Determine the <u>DNA sequences</u> (ref. 3 billion base pairs)
3 Store this information in <u>databases</u> and make the sequence totally and freely <u>accessible</u>
4 Improve tools for <u>data analysis</u>
5 Transfer related technologies to the <u>private sector</u>
6 Address the <u>ethical, legal, and social issues</u> (ELSI) that may arise from the project.

**[Benefits] [4m]**

**Molecular Medicine [Any 2 for 1m]**
1 improve diagnosis of disease (e.g. detecting rare alleles involved in common disease)
2 detect genetic predispositions to disease e.g. allow the identification of gene variants that are important for the maintenance of health, particularly in the presence of known environmental risk factors
3 facilitates the study of complex diseases involving multiple genes and their interactions with other genes, and environmental factors
4 create drugs based on molecular information
5 design “custom drugs” (pharmacogenomics) based on individual genetic profiles
6 application of HGP to gene therapy

**Comparative genomics [Any 2 for 1m]**
7 HGP data allows for comparative genomics / comparative studies with genomes of other model organisms. This is useful for development of animal models of disease development.

8 Allows quick identification of genes encoding proteins found to be involved in cancer/disease through homology searches of identified gene sequences with database sequences.

**DNA Identification [Any 2 for 1m]**
9 identify potential suspects whose DNA may match evidence left at crime scenes
10 exonerate persons wrongly accused of crimes
11 identify crime and catastrophe victims
12 establish paternity and other family relationship
13 match organ donors with recipients in transplant programs

**Bioarchaeology, Anthropology, Evolution, and Human Migration [Any 2 for 1m]**
14 Study evolution through germline mutations in lineages
15 Study migration of different population groups based on maternal inheritance
16 Study mutations on the Y chromosome to trace lineage and migration of males